# AI Messaging Sales Assistant

One AI assistant that answers customers on WhatsApp, Instagram and Messenger, books appointments, captures leads, and hands over to staff. Built for any business, with self-service sign-up.

Planning document, version 6

## 1. Goal

Answer every customer message within seconds, 24 hours a day, in Chinese and English. Capture leads, book appointments, and hand over to a person whenever one is needed.

| For | What it delivers |
|---|---|
| Customer | Instant replies in the chat app they already use, in their own language. A person is always one message away. |
| Staff | One inbox for all channels, an AI summary on every handover, and a simple way to give the chat back to the AI. |
| Business owner | A web admin panel to change prices, FAQ and AI behaviour without a developer, plus a daily summary on the phone. |
| Platform owner (you) | Businesses sign up and set up by themselves. You see usage and revenue, and can pause any business. |

## 2. How it works

```
Customer (WhatsApp / Instagram / Messenger)
            |
            v
   Chatwoot inbox  --->  n8n (routes the message)
                              |
             staff already handling?  yes -> AI stays silent
                              | no
                              v
                        AI service
                  reads this business's settings
                  uses tools: check calendar, book,
                  save lead, hand over to staff
                              |
                              v
            Reply sent back to the customer (seconds)

Needs a person? AI writes a summary note, staff are alerted,
AI stays silent until staff type /bot.
```

## 3. What the system includes

| Module | What it does |
|---|---|
| Chat | One inbox for three channels. AI answers first, staff take over at any time. Assignment, labels, private notes, canned replies. |
| Customers | Customer list with search, filters, tags, tiers, saved views, bulk actions and list settings. One customer ID across channels (matched by phone). Sub-pages: **Conversations** (every chat, including resolved, with search inside message text), **Tags** (customer tags and chat tags: create, rename, merge, colour), **Orders** (simple order or transaction records), **Review invitations** (who was invited, rating, whether they reviewed). |

| Module | What it does |
|---|---|
| Bookings | Real availability from Google Calendar, **one calendar per practitioner or resource**. Book, reschedule, cancel, reminders. |
| Marketing | Template broadcasts and reactivation, only to customers who opted in. |
| Workflow | Automations (routing, reminders, follow-ups) edited visually in n8n. |
| Reports | Chats, leads by score, bookings, handovers, reply time, top questions. |
| Team | Staff accounts, roles, automatic assignment, routing by skill. |
| Settings | AI behaviour, catalog, resources, FAQ, hours, alerts, channel connections, plan and billing. |

### Smart extras

- **AI suggested replies** for staff, sent with one click.
- **Auto summary, tags, intent and mood** on every chat.
- **Translation view** so staff read and reply across Chinese and English.
- **Click-to-chat link and QR code** that also records where each customer came from.
- **After-hours mode**, **self-service rescheduling** in chat, and a **weekly owner insights** report.
- Later: aftercare and review requests, rebooking reminders, no-show recovery, waiting list, knowledge import from a price list or web page, review queue for unsure answers, deposit links.

### More features for a complete product

| Area | Features |
|---|---|
| Safety and control | **Approval mode** (AI drafts, staff approve; switch to automatic per channel or topic once trusted). Staff timeouts and escalation (holding message, reminder, reassign, alert the owner). Business hours and holiday rules. Consent, opt-out (STOP) and a customer blocklist. |
| Booking | Buffers between appointments, cancellation policy, deposits, packages and memberships (prepaid sessions), capacity and party size, no-show tracking, **multi-branch** (own hours, calendars and staff per branch). |
| Messaging | WhatsApp template manager (create, translate, approval status). Automatic translation of staff replies. Comment-to-DM for Instagram and Facebook. Website forms and chat widget. |
| Team | Tasks from a chat with assignee and due date. Quick-reply library and media library. Fine-grained roles and permissions, tag-based auto-assignment. Customer satisfaction survey. Customer list column settings. |
| Customer data | All-conversations list with filters and full-text chat search. Tag management for customer tags and chat tags. Orders list. Review invitation list with ratings. Media library with folders. Website form submissions list. |
| Automations | A template library with editable triggers and messages: welcome, reminders, post-visit survey, rebooking, no-show recovery, birthday, review request, hot-lead follow-up. |
| Platform | Meta setup wizard, quick-start checklist, help center and in-app support chat, feature requests, data import and export, API and webhooks, status page, branding per business. |

**Later:** e-commerce and order integration, coupons and discount codes, more channels (TikTok, LINE, Telegram, email, WeChat), plugin store, security certifications, native mobile apps.

## 4. For every industry, with self-service sign-up

| Building block | How it fits any business |
|---|---|
| Universal base | Replies, FAQ answers, lead capture and scoring, handover. Works even with no booking. |
| Catalog | Services or products with name, price and description in each language. |
| Booking types | Appointment with a resource, table, viewing, class, quote visit, or none. |

| Building block | How it fits any business |
|---|---|
| Custom lead questions | The owner chooses what to ask (budget, party size, location and so on). |
| Starter templates | Clinic, dental, salon, real estate, restaurant, education, repair, retail and more. Pre-fill everything, fully editable. |
| AI-assisted setup | Drafts the setup from a description, website or price list. The owner approves before going live. |

### Sign-up journey

| | |
|---|---|
| 1 | Sign up and verify email |
| 2 | Describe the business, or paste the website or price list |
| 3 | AI drafts the setup, owner reviews and approves |
| 4 | Connect Google Calendars and WhatsApp with guided steps and test buttons |
| 5 | Test chat, choose a plan, go live |

## 5. Parts and tools

| Part | Tool | Role |
|---|---|---|
| Channels | Meta WhatsApp, Instagram, Messenger APIs | Where customers write. Each business uses its own number and Meta account. |
| Inbox | Chatwoot (open source) | Staff inbox. One account per business, created automatically. |
| Automation | n8n (open source) | Routes messages, sends replies, handover alerts, reminders, reports. |
| AI service | TypeScript API service in the aichat repo + Claude | Prompt, business knowledge and tools. Replies in the customer's language. |
| Admin panel | Next.js web app (React) | Business settings, customers, bookings, reports. Platform-owner console. Theme colour and logo per business. |
| Calendar | Google Calendar API | Availability and bookings, one calendar per practitioner. |
| Database | PostgreSQL | All data, separated per business with row-level rules. |
| Sign-up and billing | Public site + Stripe or local provider | Registration, plans, usage limits, invoices. |

## 6. Technical stack

All of our own code is TypeScript, in one repository, for both the web app and the server. Chatwoot and n8n run as ready-made Docker images, so they need no custom programming, only configuration.

### Languages and frameworks

| Part | Technology | Notes |
|---|---|---|
| Language and tooling | TypeScript (strict mode), Node.js 22, pnpm monorepo, ESLint | One codebase with shared types between the web app and the API. |
| API, AI and workers | Fastify (API), Zod (validation), BullMQ (background jobs on Redis) | Webhooks from Chatwoot and Meta, AI tool calls, booking, usage limits, reminders, billing events. |
| AI model | Claude through the Anthropic TypeScript SDK | Default model Claude Haiku 5.5, set by one setting and changeable per business or plan. Tool use for calendar, booking, leads, handover. |

| Part | Technology | Notes |
|---|---|---|
| Web app (admin, sign-up, platform console) | Next.js, React, Tailwind CSS, shadcn/ui components | Polished, interactive screens. Colours are theme variables, so each business can pick its own colour and logo, plus dark mode. |
| Inbox | Chatwoot (Ruby on Rails, Docker image) | Configured through its API and webhooks. One account per business, created through its platform interface. |
| Automation | n8n (Node.js, Docker image) | Workflows stored as JSON files in the repo and imported on start. |
| Calendar | Google Calendar API, googleapis library | Service account. Each business shares its calendars with it. |
| Billing | Stripe API (or a local provider) | Subscriptions, usage limits, invoices, webhooks. |

### Data, infrastructure and operations

| Area | Technology | Notes |
|---|---|---|
| Database | PostgreSQL 16 with Drizzle ORM | SQL-first, works well with row-level security keyed by business ID. Separate databases for Chatwoot, n8n and our service on one server. |
| Cache and queues | Redis | Needed by Chatwoot. Also used for rate limits and background jobs. |
| Packaging | Docker Compose | Whole stack in one file, easy to move between servers. |
| Web server and HTTPS | Caddy | Automatic certificates. Public address needed for Meta webhooks, through a tunnel such as ngrok during development. |
| Backups | Scheduled pg_dump to cloud storage | Per-business export and deletion available. |
| Monitoring | Uptime Kuma, health checks in the admin dashboard | Alerts to Telegram or email. |
| Email | An email-sending service over SMTP | Verification emails and notifications. |

### Security and quality

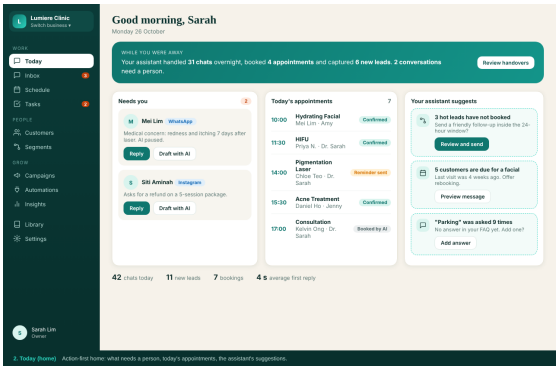
| Area | Approach |
|---|---|
| Login | Email and password with argon2 hashing, secure sessions, optional two-step login (TOTP) for owners. |
| Secrets | Meta tokens and Google credentials encrypted at rest. Secrets kept in environment files, never in git. |
| Isolation | Business ID on every table plus row-level security. Automated tests try to read across businesses. |
| Audit | Sign-ins, support access, pauses and deletions are logged. |
| Testing | Vitest for logic (booking slots, handover rules, limits), Playwright for end-to-end screens, sample-message tests for workflows. GitHub Actions runs checks on every change. |
| Repository layout | aichat repo: apps/web, apps/api, shared packages, Docker Compose, database migrations, n8n workflow JSON, templates, docs. |

**Decided:** TypeScript for all our own code. **Choices to confirm:** login library (Auth.js or Better Auth), Stripe or a local payment provider, and the email-sending service.

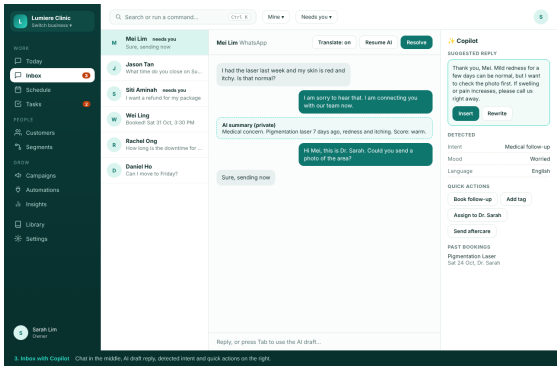
# 7. Design

A calm teal, AI-first workspace, original to this product. Six of the most important screens are shown below. All 14 screens are in the separate page-layouts PDF.

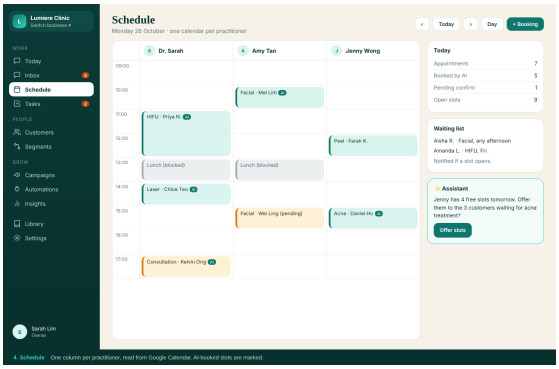
| Element | Choice |
|---|---|
| Principles | Action first (home shows what needs a person), AI everywhere (drafts, summaries, next steps), cards over tables, plain wording. |
| Colours | Deep teal #08302d menu, teal #0f766e primary, mint #2dd4bf accent, cream #f7f5ef background. Status colours: hot, warm, cold. |
| Type and shape | Clean sans-serif interface, serif page titles, rounded cards, pill tabs. |
| Menu | WORK: Today, Inbox, Schedule, Tasks. PEOPLE: Customers (list, conversations, tags, orders, review invitations), Segments. GROW: Campaigns, Automations, Insights. Then Library and Settings. |
| Themes | Teal by default. Each business can pick its own accent colour and logo, and use dark mode. |



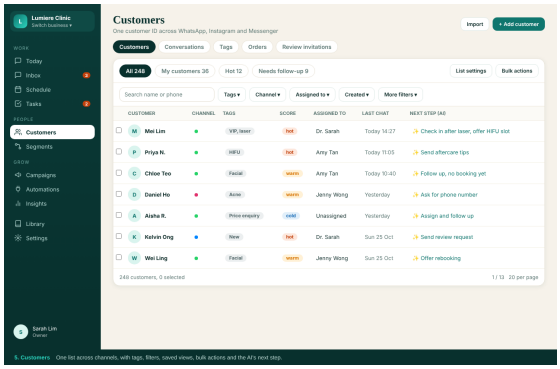

Today: what needs a person, today's appointments, assistant suggestions



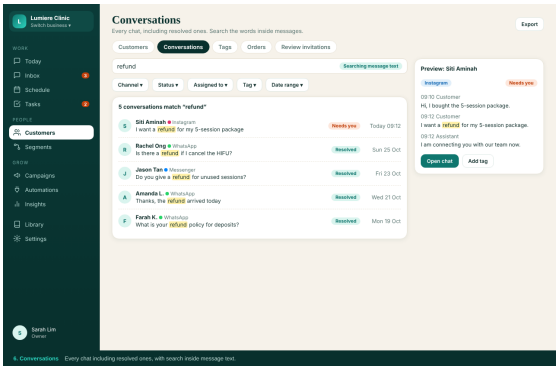

Inbox with Copilot: draft reply, intent and quick actions



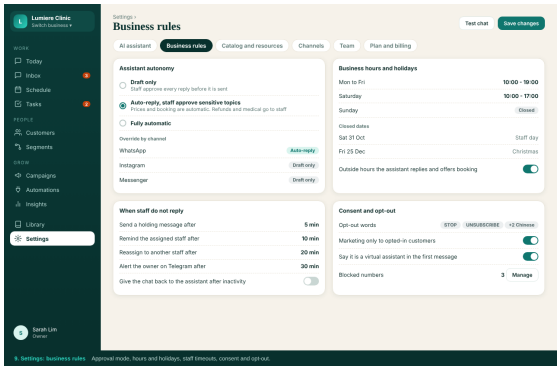

Schedule: one Google Calendar per practitioner, AI-booked slots marked




Customers: tags, filters, saved views, bulk actions, AI next step




Conversations: all chats with search inside message text




Business rules: approval mode, hours, staff timeouts, opt-out

## 8. Handover, safety and separation

### The AI hands over to staff when

- The customer asks for a person.
- Complaint, anger, refund or payment problem.
- Medical, legal or safety concern. The AI never gives such advice.
- The question is outside the business's knowledge, or the AI is unsure.

### Safety and separation

- Answers only from the business's own knowledge. Never invents prices, offers or availability.
- Tells customers it is a virtual assistant. Customer text is treated as untrusted.
- Every record carries a business ID, and database rules block cross-business reads.
- Encrypted credentials, two-step login for owners, audit logs, data export and deletion.
- Sign-up protection: email verification, bot check, limits, and a kill switch per business.
- If the AI service fails, the customer gets a polite message and staff are alerted.

## 9. Plans, hosting and costs

| Item | Approach |
|---|---|
| Plans | Free trial, then paid monthly plans with limits on AI replies, staff and resources. At the limit the AI pauses, the owner is told, and messages still reach the inbox. |
| Hosting | Free hosting for development, the demo and an invite-only beta. Public sign-up needs a stable paid server with backups. |
| AI cost | Pay per use and small (Claude Haiku 5.5 by default). Counted per business and passed on through plans. |
| Meta fees | Billed by Meta directly to each business. |
| Other | Domain, email sending, and payment provider fees. |

## 10. Free-first testing and build phases

Almost everything can be built and tested for free. The AI is the only paid part, and testing it costs cents.

| Part | How to test it for free |
|---|---|
| Servers | Docker on your own computer. No paid hosting while building. |
| Public address | A free tunnel (ngrok free account or Cloudflare quick tunnel) so Meta can reach your computer. |
| First chat channel | Chatwoot website chat widget in your browser. No Meta account needed. |
| WhatsApp | Meta's developer test number, to a few approved numbers, without business verification. Check Meta's current rules. |
| Instagram and Messenger | A Meta app in development mode works for people with a role on the app, so no app review while testing. |
| Google Calendar | Free API, a test calendar on a free Google account. |
| AI | A fake AI provider with scripted replies runs all automated tests for free. Real Claude is used for a few manual tests (about 100 test chats is roughly $0.50, my estimate). Needs an Anthropic API key with a little credit. |

## Local-first setup

Decision: build and test on localhost (your own computer) first. Going live is far away, so no server is needed yet.

| Item | Plan |
|---|---|
| Runs on | Docker Desktop on your computer, 8 GB RAM recommended. One command starts the whole stack. |
| Local addresses (planned) | Admin web app localhost:3000, API localhost:4000, inbox localhost:3001, workflow editor localhost:5678. |
| Meta and Stripe | Free tunnel to localhost: ngrok free fixed address or Tailscale Funnel (stable), or Cloudflare quick tunnel (changes on restart). Stripe's command-line tool forwards test payments. |
| Limits and moving | The computer must stay on and customers cannot reach localhost. For 24/7 testing, an Oracle Cloud Always Free server is an option (card check, availability varies). Paid hosting only at go-live (phase 7); the Docker stack moves as is. |
| How to run | Install Docker Desktop, clone the repo, copy .env.example to .env, run docker compose up, open the admin address. |

## Build phases

| Phase | What gets built | How it is tested | Cost |
|---|---|---|---|
| 0. Test base | Docker stack on your computer (Chatwoot, n8n, PostgreSQL, Redis), tunnel, business ID on every table, fake AI provider, website chat widget. | A message in the web widget gets a scripted reply and shows in the inbox. | Free |
| 1. Core assistant | Real Claude behind the provider setting. Bilingual answers from catalog and FAQ, lead capture and scoring, handover with private note and alert, approval mode and autonomy levels, business hours, staff timeouts, consent and opt-out. | Scripted English and Chinese test chats. A few real-AI chats to check quality. | Cents of AI |
| 2. Booking | Google Calendar availability, booking and cancel, reminders, buffers, cancellation policy, deposits, capacity. | Book on a test calendar. A second booking for the same slot must fail. | Free |
| 3. Business admin panel | Today, Inbox with Copilot, Schedule, Customers (one ID per person), all conversations with full-text search, tag management, tasks, library (replies, media folders, forms), settings, roles and permissions. | Change a price in settings and see the assistant use it. Test chat for every change. | Free |
| 4. Real channels | WhatsApp test number, Instagram and Messenger in development mode, template manager, Meta setup wizard. | Real messages with a few approved numbers. | Free |
| 5. Automations and marketing | Automation templates, reminders with WhatsApp templates, satisfaction survey, rebooking, review request, opt-in broadcasts, review invitation list, orders list, reports, comment-to-DM. | Run each automation on test customers. | Free |
| 6. Many businesses and self-service | Tenant isolation, sign-up and email verification, AI-assisted setup, onboarding, plans and limits, billing, platform-owner console, abuse protection, legal pages. | Two test businesses cannot see each other's data. Full sign-up journey works. | Free (payment tests use the provider's test mode) |
| 7. Beta readiness | Meta business verification, real WhatsApp number, paid server with backups and monitoring, domain, email sending, legal review. Invite-only beta with a few businesses. | Real customers in a small, watched group. | Paid hosting starts here |
| 8. Public launch and growth | Open sign-up. Multi-branch, packages and memberships, e-commerce, more channels, remaining extras. | Gradual rollout with usage limits. | Paid |

Phases 0 to 5 use free tools apart from small AI test costs. Paid hosting begins at phase 7, when real customers arrive.